MACRO L3 - 2014-2015

Lecture 6

F. PORTIER

Session 8

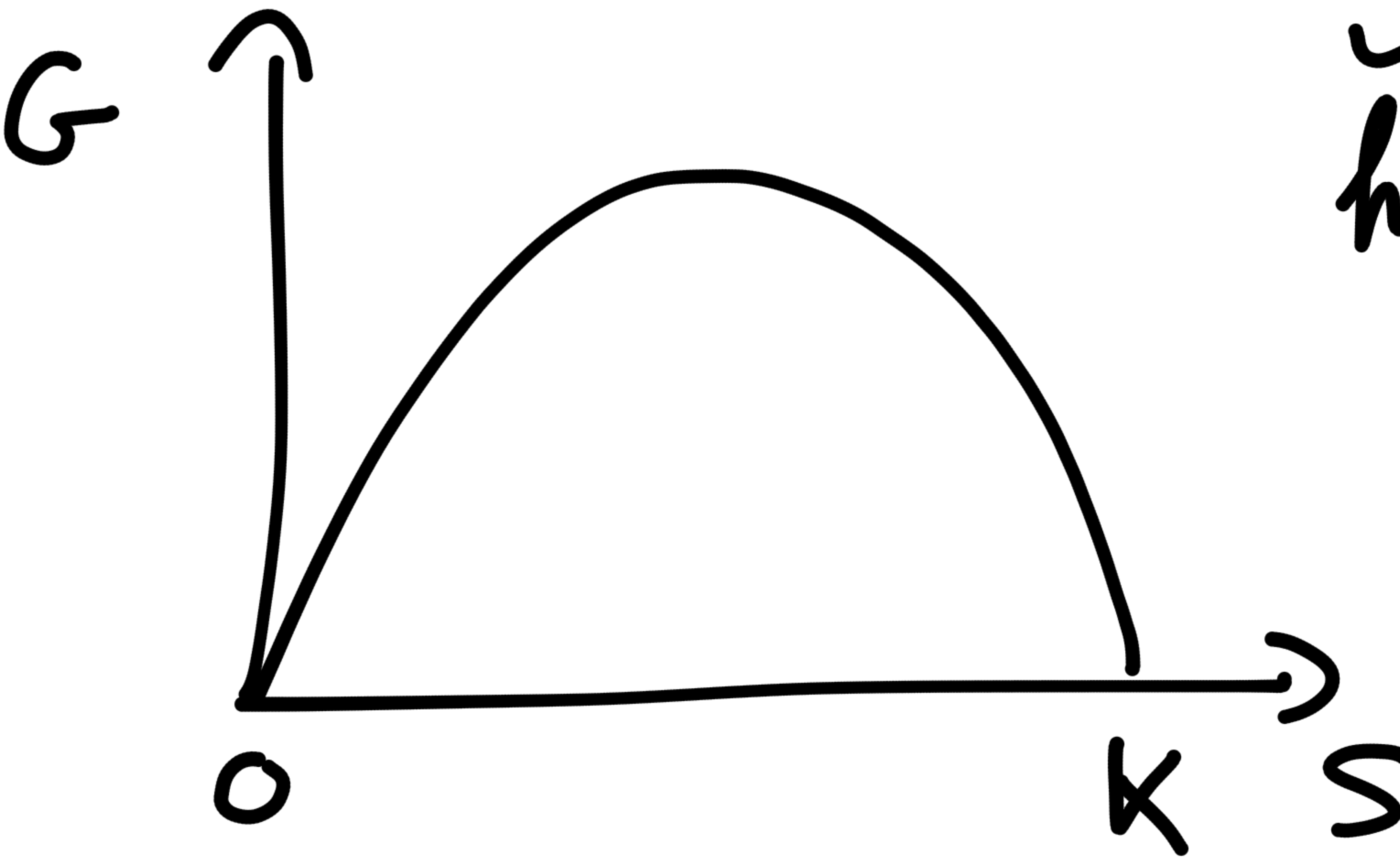
These notes are those I wrote during class. They are posted "for the record".

Session 8

Nov 24, 2014

Biology

$$\dot{S} = G - \underbrace{H}_{\text{harvest}}$$



We have derived from human behavior

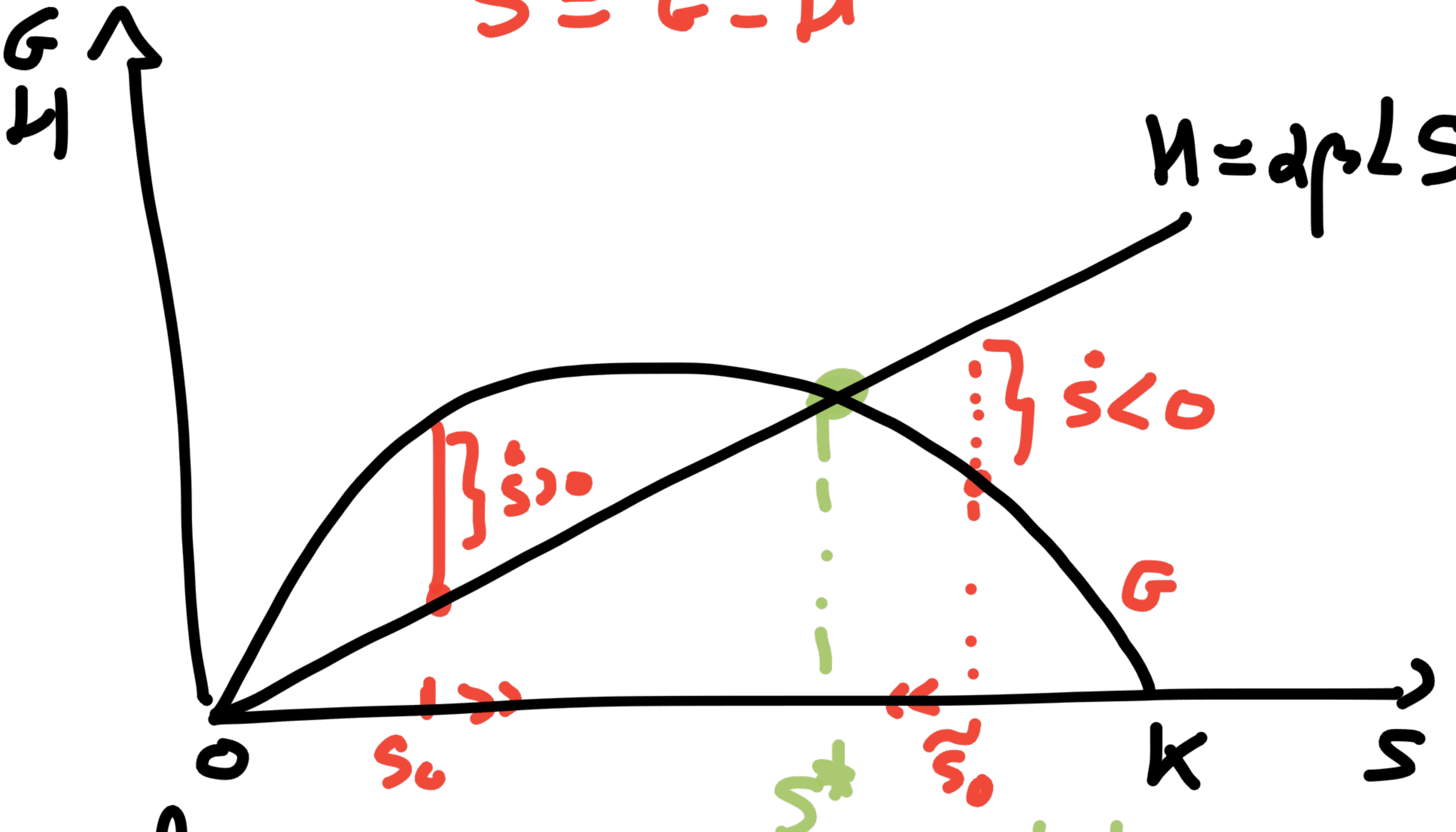
$$H(t) = \alpha \beta S(t) L(t)$$

↳ Dynamics of the Forest

$$\boxed{\dot{S}(t) = r S(t)\left(1 - \frac{S(t)}{K}\right) - \alpha\beta S(t) L(t)}$$

for given population $L(t)$

$$\dot{S} = G - H$$



for given $L(t)$

s^* = stationary level given L

$$\dot{s}^* = G(s^*) - H(s^*) = 0$$

Demographics

- mortality rate : constant d
- fertility rate : variable: $b + f(t)$

$$\boxed{f(t) = \frac{\phi H(t)}{L(t)}}$$

$$\frac{H(t)}{L(t)} = \text{harvest/person}$$

$$\dot{L}(t) = \left[b - d + f(t) \right] L(t)$$

- Assume: $b - d < 0$

  meaning: if $S = 0$, then $H = 0$ $f = 0$

  and $\frac{\dot{L}}{L} < 0$ and $\lim_{t \to \infty} L(t) = 0$

therefore

$$\dot{L}(t) = \left(b - d + \phi \frac{M(t)}{L(t)}\right) L(t)$$

and $M(t) = \alpha \beta L(t) S(t)$

such that

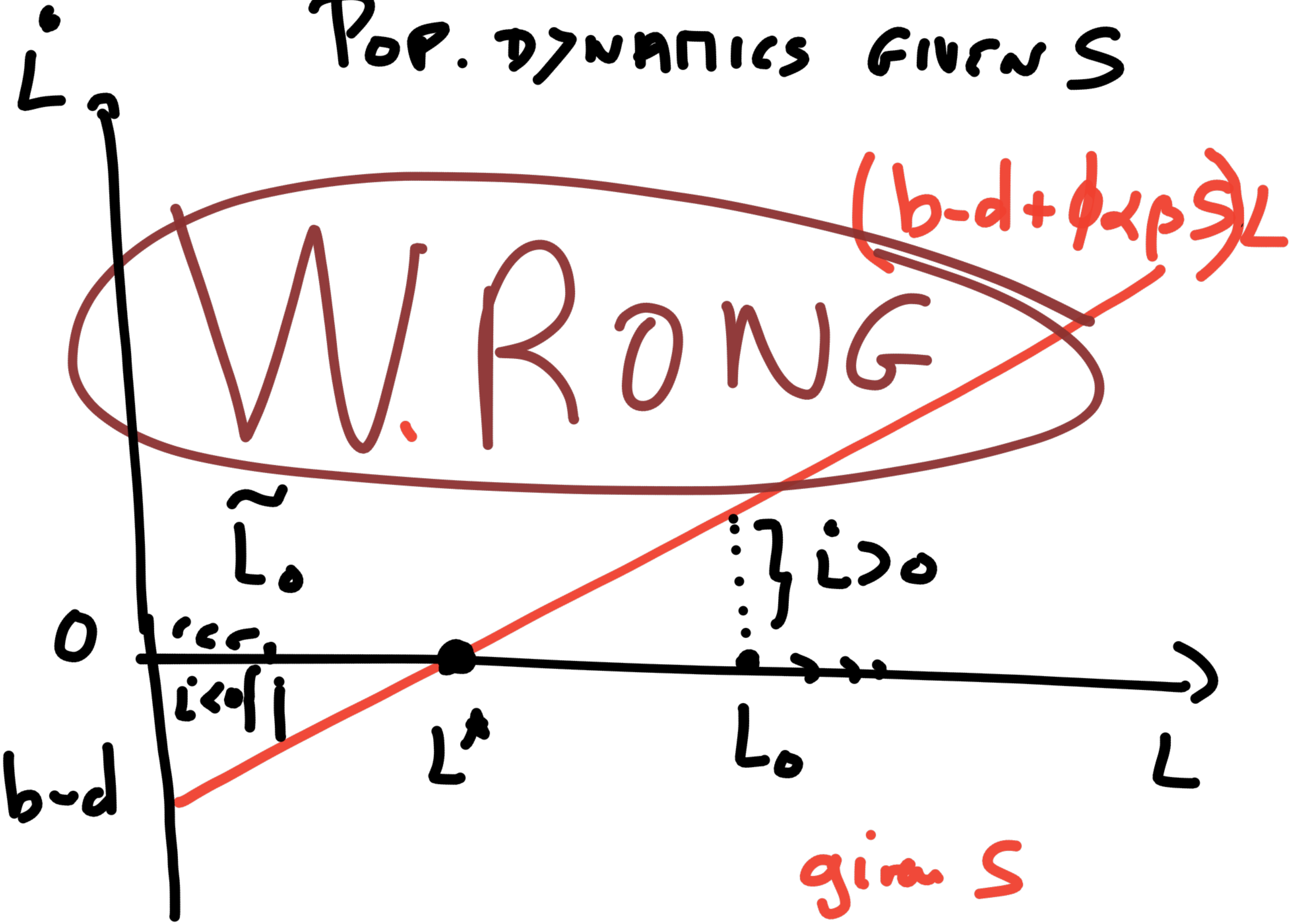
$$\boxed{\dot{L}(t) = \left[b - d + \phi \alpha \beta S(t)\right] L(t)}$$

POPULATION DYNAMICS given S

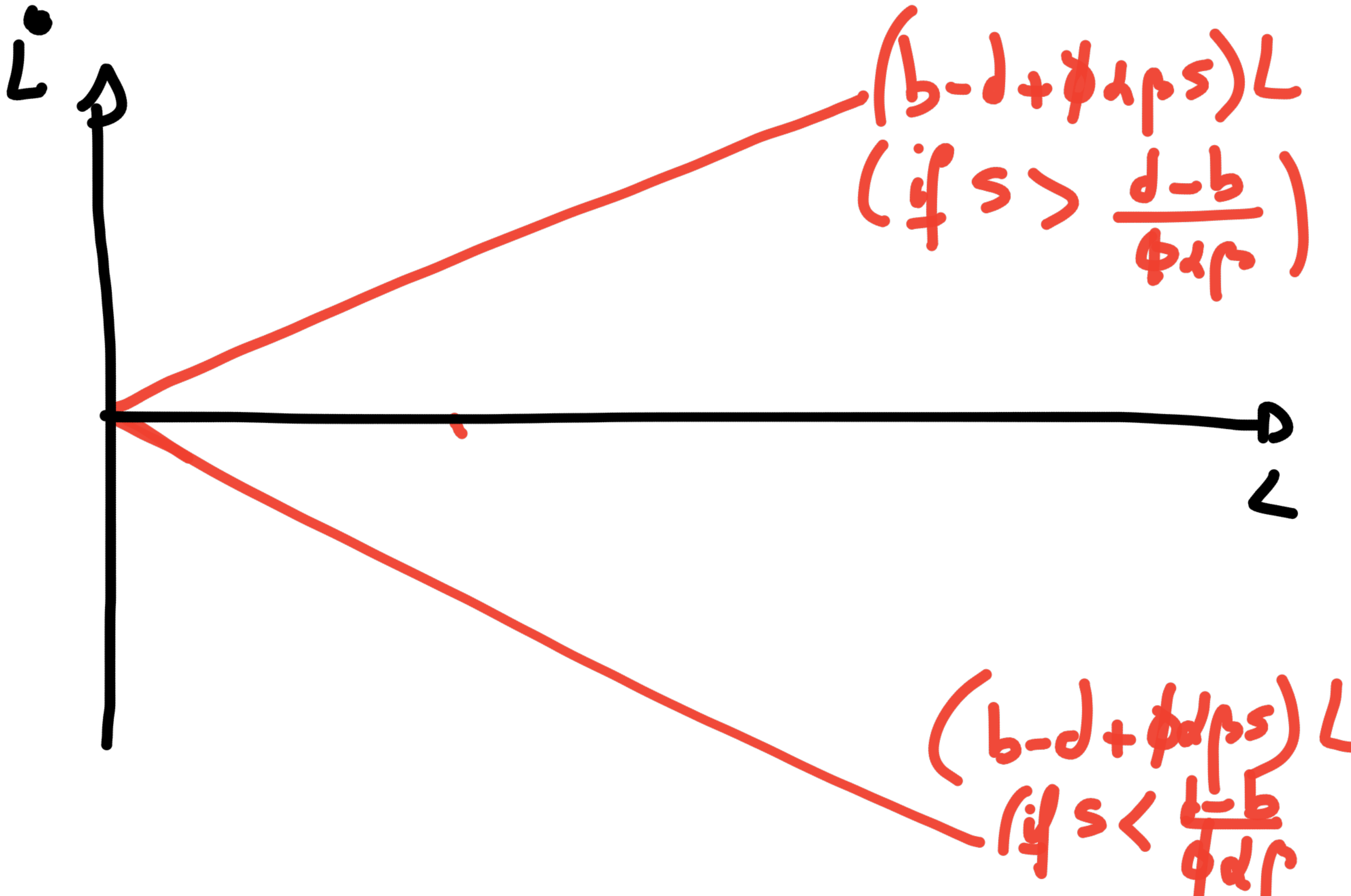
WARNING

The following figure that I drew in class in wrong - The correct figure is displayed below.

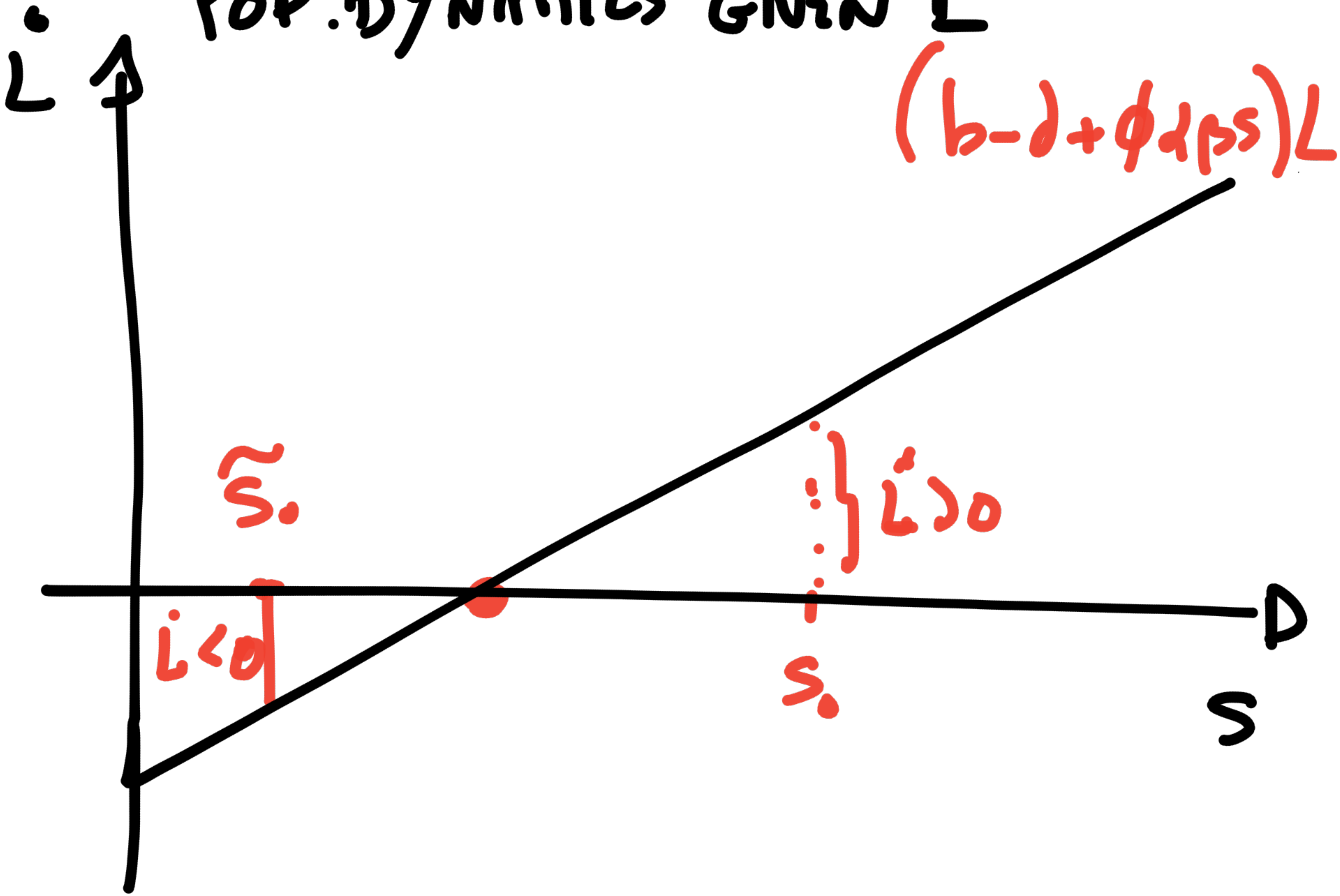
Pop. dynamics given S



POPULATION DYNAMICS GIVEN S



POP. DYNAMICS GIVEN L



ISLAND'S DYNAMICS

$$\begin{cases} \dot{S}(t) = r S(t)\left(1 - \frac{S(t)}{K}\right) - \alpha \beta S(t) L(t) & \text{(A)} \\ \dot{L}(t) = \left[b - d + \phi \alpha \beta S(t)\right] L(t) & \text{(B)} \end{cases}$$

$\hookrightarrow$ The model is solved!

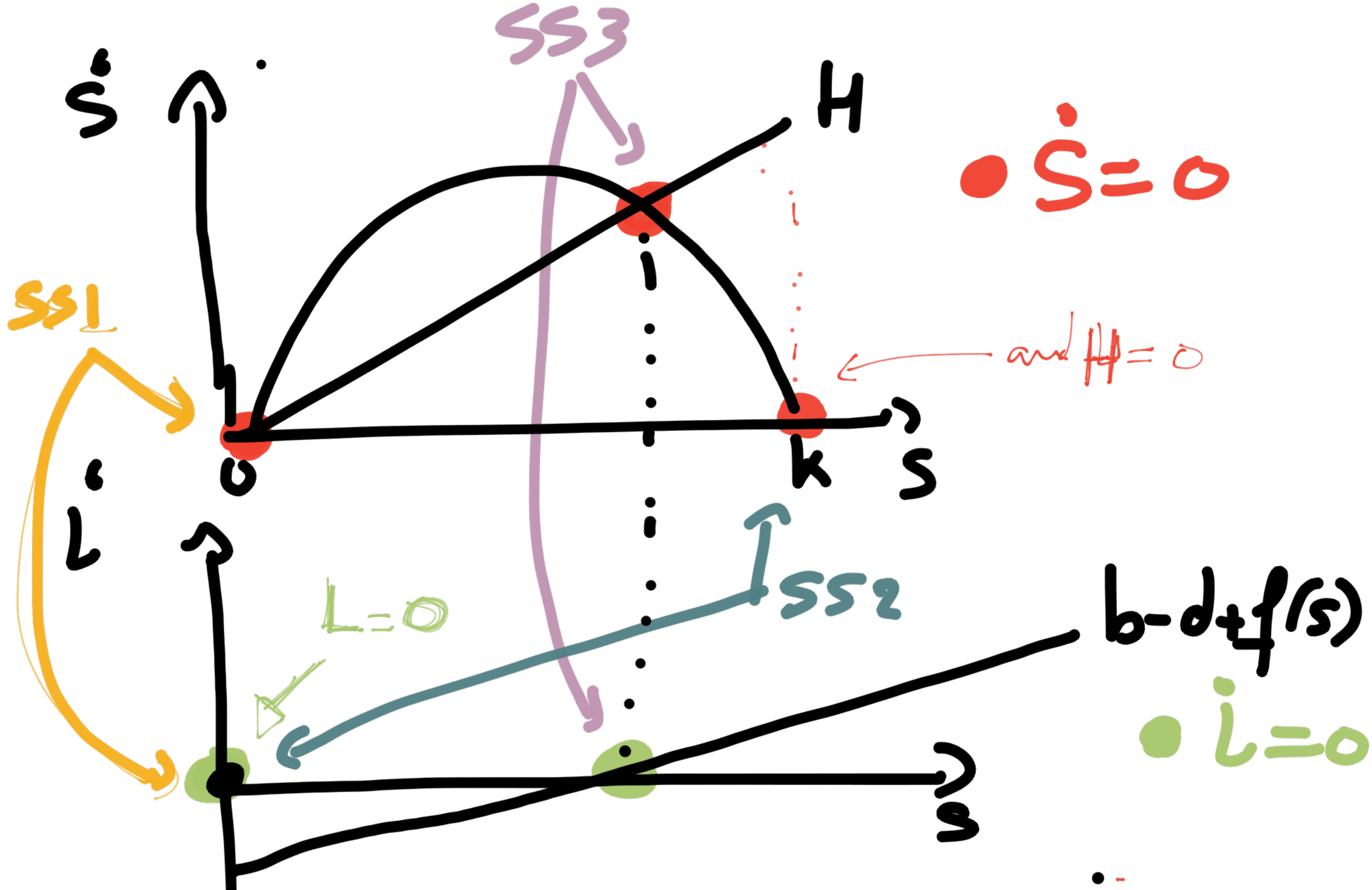
initial conditions

$$\boxed{S_0, L_0} \to (\dot{S}_0, \dot{L}_0) \to (S_1, L_1) \to (\dot{S}_1, \dot{L}_1)$$

etc. . . . . .

3 Steady States

$\dot{i} = 0 \quad \dot{s} = 0$



<u>SS1</u> : $L=0, S=\infty$

Desertic island

<u>SS2</u> : $L=0, S=K$

Unoccupied island
(Virgin Island)

<u>SS3</u> $L>0 \quad S>0$

Sustainable island

Mathematically

$$\begin{cases} \dot S = 0 \Rightarrow rS\left(1-\frac{S}{K}\right) = \alpha\beta S L \quad (A) \\ \dot L = 0 \Rightarrow (b-d+\phi\alpha\beta S)\times L = 0 \quad (B) \end{cases}$$

(B) 2 sol $\rightarrow L = 0$

$\searrow S = \dfrac{d-b}{\phi\alpha\beta}$

(A) : if $L = 0$, 2 sol $\rightarrow S = 0$

$\searrow S = K$

if $S = \dfrac{d-b}{\phi\alpha\beta}$, $L = \dots\dots$

If $s = \dfrac{d-b}{\phi \alpha \beta}$, then

(A) $\Rightarrow L = \dfrac{r}{\alpha \beta}\left(1 - \dfrac{d-b}{\phi \alpha \beta k}\right)$

$\hookrightarrow$ this is the interior SS

= sustainable island